Design review · status after V8.1

Taif / Palm House · the review of 30 Oct 2026 (`docs/review/design_review.typ` → `output/Design_Review_V6.pdf`, tag `review-1`, written against V6.2) · status on 1 Nov 2026 after V6.3, V7.0, V7.1, V8.0 (fire safety, courtyard layout, air system, south wing v2) and V8.1 (a checkpoint: the roof ladder, the lowered roofline, the office from the suite court, one site entry); the rows updated in V8.1: E-08, F-07, T-03, E-04, E-06, S-05 · every critical and major finding: 30 in all, **resolved** 12 · **partly** 11 · **open** 7

Id	Sev.	Where	Status	Version	What was done · what is still open
Y-01	critical	Courtyard · the envelope	**resolved**	V7.0 / V7.1	Enclosure option A: W1 on the gallery edge, W2 on the family living's edge, W3 over the living's courtyard glass, black steel divided lites, the projection 1.55 m (B3 / R11 / C21–C23 at x 5.80); the courtyard open to the sky. V7.1: semicircular heads over the three-arch arcade.
A-01	critical	Ageing · the lift	**resolved**	V6.3	The lift installed at construction, two stops; the shaft a placeholder until a supplier's drawing (open item 15: headroom ≤ 3.40 m / ≤ 3.70 m with a raised head).
S-03	critical	Site · north and lane setbacks	**partly**	V6.3	The house 0.30 m south: 0.20 m to the lane setback line, 0.80 m to the north street line, provisional. **Open:** the legal survey (open item 1).
E-08	major	Services · fire and life safety	**resolved**	V8.0 / V8.1	Home-level strategy (owner): interconnected mains smoke alarms with battery back-up in every bedroom and on the routes, heat, CO and gas detectors, extinguishers and a blanket; escape casements without screens in every sleeping room (the maid room's at sill +1.10); thumb-turns on the exits (`studies/fire_escape.typ`). V8.1: the first floor's exits are the main stair plus the escape window of every sleeping room; the suite court is a fresh-air area, not an exit; the roof ladder is no escape device.
G-01	major	GF · living wall (west)	**resolved**	V6.3	The solid TV wall with the entrance arch and the arched TV niche.
G-04	major	GF · maid room	**resolved**	V7.0	The kitchen's north wall at y 7.0: maid room 7.28 m² clear, 2.49 m at the bed.
F-01	major	FF · gallery width	**resolved**	V7.0	The projection 1.55 m: the gallery 1.405 m clear, 1.235 m at the column sleeves.
F-02	major	FF · door widths	**resolved**	V6.3	Bedroom and bathroom doors 1.00 m openings; exceptions recorded.
F-06	major	FF · family living	**resolved**	V7.0	W2 on the courtyard edge: a conditioned room.
F-07	major	FF / roof · roof access	**resolved**	V7.0 / V8.1	V8.1: the roof by a manufactured maintenance ladder in a 0.90 × 0.30 recess of a stone pilaster on the gym's east wall in the suite court (3.70 m, the stiles 1.10 m above the roof, a self-closing gate in the court guard, a fall-arrest rail), behind a locked walnut panel (`studies/roof.typ` p. 2); the guarding continuous to every piece of equipment; the internal stair stops at the first floor, the roof slab closed over the well. **Open:** the fall protection confirmed by the specialist (open item 4).
X-01	major	Circulation · serving the majlis	**open**		The food route from the kitchen to the men's majlis still crosses the dining and the riwaq (open item 19): an outside service path to a majlis serving door, or accept.
G-02	major	GF · living + dining	**partly**	V7.0	The courtyard well decided; V8.0: the clerestories get the winter noon sun on their whole height (the suite court open to the south facade). **Open:** the daylight simulation (open item 16) and the garden door (S-04).
G-03	major	GF · kitchen	**partly**	V8.0	The hood's make-up air: a filtered motorised inlet under the stone band on the east wall, interlocked with the hood (air v3.1). **Open:** the MEP sizing (open item 9).
G-07	major	GF · bathrooms	**partly**	V6.3	The first-floor showers reachable by the lift. **Open:** a ground-floor shower.
Y-02	major	Courtyard · size for daily use	**resolved**	V8.0	The courtyard layout (owner): the seating one step down with three benches for six to eight and a low table; the TV wall; children's play in the garden, not in the courtyard; a hose point at the enclosure (`studies/courtyard_layout.typ`).
Y-03	major	Courtyard · sunken seating safety	**resolved**	V8.0	The 0.45 m pit replaced by one 0.15 m step with a honey-stone nosing and a riser light; level strips ≥ 0.6 m at every glass line; the drain and an independent overflow (the drainage section).
T-01	major	Structure · X5	**partly**	V8.0	Load path revised, **improved, not verified**: R6 removed (the suite court open to the south facade); the gym's east wall on B19 between C19 and C20, the double wall on B8 and a new B22, the office wall non-load-bearing on rib B21. **Open:** the engineer confirms (open item 3).
T-03	major	Structure · soil and foundations	**open**		No soil report (open item 7): the pads, the lift pit and the breeze channel are assumed.
E-01	major	Services · the malqaf's value	**resolved**	V8.0	Air system v3 / v3.1 (owner): one tower with two separate paths: a gentle courtyard breeze (a damper, a shallow channel in the slab, the grille under the TV, an assist fan) and the fresh-air intake of the HRV in the tower's base on the roof; no water, no pit, no entry (`studies/air_system.typ`). The MEP engineer sizes it (open item 9).
E-02	major	Services · ventilation and dust	**partly**	V8.0	A 750 m³/h class HRV with an ISO ePM1 ≥ 50 % filter, continuous + boost, a dust mode, the kitchen make-up air (all assumed). **Open:** the MEP design.
E-03	major	Services · plumbing	**partly**	V8.0	The courtyard's drainage section (pit drain, overflow, pipes east) and the suite court's drains (linear drains, box gutter, overflow) drawn. **Open:** the plumbing study (open item 10).
E-04	major	Services · solar PV provision	**partly**	V8.1	PV zones inside the guarded zones: 9.8 m² on the gym roof and the office roof (inside the set-back guardrail, outside the clerestories' south sector) and 20.5

					m^2 on the north block. **Open:** the inverter / battery room, the conduits (the site and services plan); more PV area if wanted.
E-05	major	Services · generator and essentials circuit	**open**		The generator point is only proposed (air study distances): the site and services plan (open item 20).
E-06	major	Services · security and the safe room	**partly**	V8.1	No safe room (owner, R-33); one site entry, the farm's 4 m main gate (no gate on the lane); office visitors met in the men's majlis, the first floor closed to them; the roof ladder behind a locked panel. **Open:** cameras, the alarm and intercoms with the site plan.
E-07	major	Services · acoustics of the living	**open**		The double-height living's finishes and absorption at the detail round.
S-01	major	Site · parking	**open**		The site and services plan (open item 20).
S-02	major	Site · guest approach	**open**		The site and services plan.
S-04	major	Site · garden access	**open**		A garden door from the living / dining: the site and services plan.
S-05	major	Site · east strip, service yard	**partly**	V8.1	The service yard screened (bins, gas); the outdoor counter and sink against the east wall on the service-door path, the grill and tannour apart in the garden (3.6 m from the house); no gate on the lane. **Open:** the full site plan.
S-08	major	Site · tanks, septic, plant	**partly**	V8.0	On the roof: the HRV in the malqaf's base, the AC units south of the tower, the tank zone, the PV. **Open:** the ground tank, septic, generator on the site plan.

The next open items, in this order

1. **The internal layout round:** the stair flip (`studies/stair_flip.typ`: study done, the owner's decision pending); bedroom 1's bathroom route (F-05, open item 17); a ground-floor shower (G-07); the food route from the kitchen to the men's majlis (X-01, open item 19).
2. **The suite court's south side and the clerestory glazing** (open items 18, 14).
3. **The site and services plan** (S-01 … S-08, E-04 … E-06; open item 20): parking, the guest approach, the garden door, the service yard, water tanks, septic, gas, the generator, the PV conduits, security, Civil Defense access.
4. **The engineers:** structure X5 and the soil report (T-01, T-03; open items 3, 4, 7); MEP sizing of the air system, the HRV and the plumbing (E-01 … E-03; open items 9, 10).
5. **The legal survey** (S-03, open item 1).
6. **The living's acoustics and the facades** at the detail stage (E-07).

The full verification and validation review is planned for V9.0, after the internal layout round. This page is a status update, not a new review.

Minor findings, suggestions and strengths are unchanged in the review; of the minors, T-02 (X3) is resolved (V6.3), F-09 (the gym's treadmill clearance) is resolved in V8.0 (the manufacturer-type zone on the west band, vibration with the engineer), Y-04 (courtyard dust and water) is answered by the drainage section and the hose point, F-05 (bedroom 1's bath) is the first next item. Nothing here is for construction: concept design, all sizes assumed until the engineers confirm.